

Two Degree of Freedom MSD Systems

EGR 345

GVSU

Outline

- quick review
- big picture and plans
- two DOF MSD systems

Quick Survey

<https://etc.ch/Uzzm>

Click for QR Code

Big Picture from Lecture 1

- what was the plan I laid out?
- how are we doing?

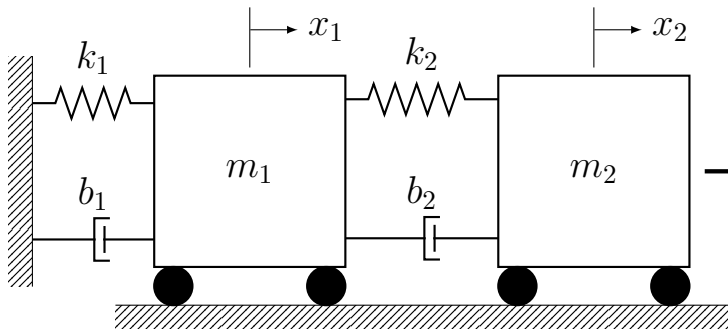
Course Plans

- two DOF MSD (today)
- pole locations (Thursday and always)
- kick-off semester projects (lab this week)
- start Bode (lecture next week)
- pendulum Bode system ID (lab next week)
- beam Bode system ID (lab soon)
- intro to feedback control
- vibration suppression control

Two Degree of Freedom MSD Systems

- how do we find the transfer function(s) for this system?

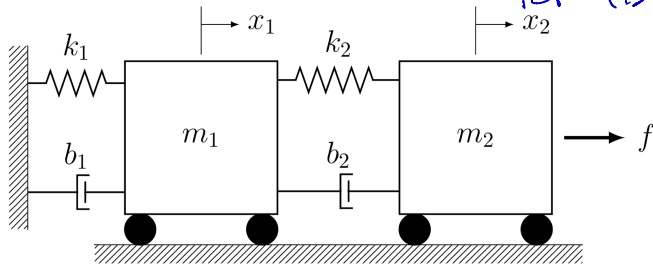
- how many TFs do we need? $\rightarrow$ a TF has one input and one output
 $\therefore$ we need 2



$$G_1 = \frac{x_1}{F}$$

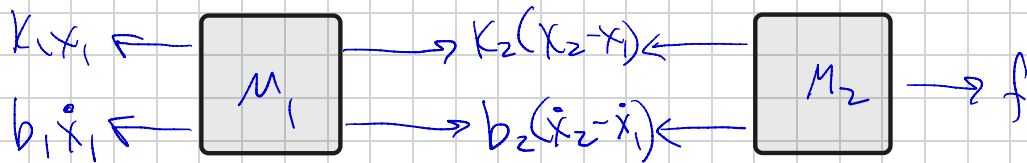
$$G_2 = \frac{x_2}{F}$$

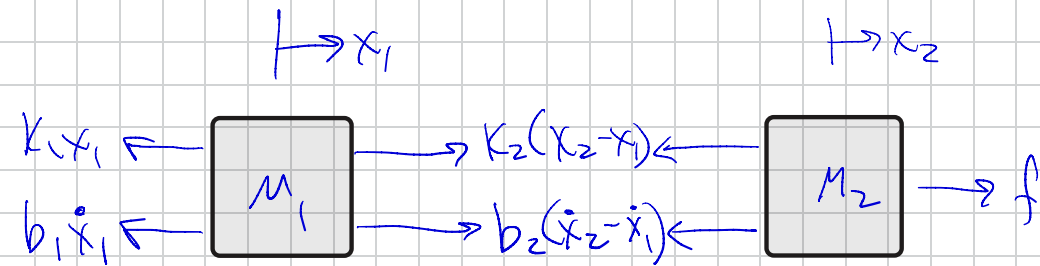
-FBD
-Laplace
-EOM



For FBD, Assume $x_2 > x_1 > 0$
 $\dot{x}_2 > \dot{x}_1 > 0$

$\rightarrow x_2$





for M_1 :

$$M_1 \ddot{x}_1 = \underbrace{K_2(x_2 - x_1)}_{\uparrow} + \underbrace{b_2(\dot{x}_2 - \dot{x}_1)}_{\uparrow} - \underbrace{K_1 x_1}_{\uparrow} - \underbrace{b_1 \dot{x}_1}_{\uparrow}$$

$$M_1 \ddot{x}_1 + b_1 \dot{x}_1 + b_2(\dot{x}_1 - \dot{x}_2) + k_1 x_1 + k_2(x_1 - x_2) = 0$$

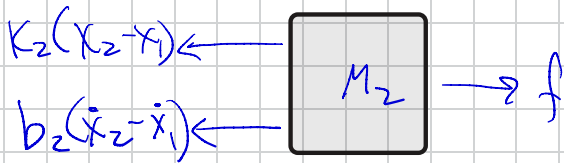
$$m_1 \ddot{x}_1 + \underline{b_1} \dot{x}_1 + \underline{b_2} (\dot{x}_1 - \dot{x}_2) + \underline{k_1} x_1 + \underline{k_2} (\underline{x_1} - x_2) = 0$$

$$m_1 \ddot{x}_1 + (b_1 + b_2) \dot{x}_1 + (k_1 + k_2) x_1 - b_2 \dot{x}_2 - k_2 x_2 = 0$$

Sign checks

- all $x_1, \dot{x}_1$, and $\ddot{x}_1$ terms are + in the m_1 equation
- all x_2 and $\dot{x}_2$ terms in m_1 equation are negative

$\rightarrow x_2$



For m_2 :

$$m_2 \ddot{x}_2 = f - \underset{\uparrow}{k_2}(x_2 - x_1) - \underset{\uparrow}{b_2}(\dot{x}_2 - \dot{x}_1)$$

$$m_2 \ddot{x}_2 + b_2(\dot{x}_2 - \dot{x}_1) + k_2(x_2 - x_1) = f$$

$$m_2 \ddot{x}_2 + b_2 (\ddot{x}_2 - \dot{x}_1) + k_2 (x_2 - x_1) = f$$

$$m_2 \ddot{x}_2 + b_2 \dot{x}_2 + k_2 x_2 - b_2 \dot{x}_1 - k_2 x_1 = f$$

all $x_2, \dot{x}_2, \ddot{x}_2$ terms in
 m_2 eqn are +

$x_1, \dot{x}_1$ terms in m_2
 eqn are negative
 → and correspond to
 x_2 and $\dot{x}_2$ terms in
 m_1 equation

$$m_1 \ddot{x}_1 + (b_1 + b_2) \dot{x}_1 + (k_1 + k_2) x_1 - b_2 \dot{x}_2 - k_2 x_2 = 0$$

$$m_2 \ddot{x}_2 + b_2 \dot{x}_2 + k_2 x_2 - b_2 \dot{x}_1 - k_2 x_1 = f$$

2 ↓ assume zero initial conditions

$$[m_1 s^2 + (b_1 + b_2)s + k_1 + k_2] x_1 - (b_2 s + k_2) x_2 = 0$$

$$[m_2 s^2 + b_2 s + k_2] x_2 - (b_2 s + k_2) x_1 = F$$

$$D_1 \quad N \quad \left[m_1 s^2 + (b_1 + b_2)s + k_1 + k_2 \right] x_1 - (b_2 s + k_2) x_2 = 0$$

$$D_2 \quad \left[m_2 s^2 + b_2 s + k_2 \right] x_2 - (b_2 s + k_2) x_1 = F$$

$$G_1 = \frac{x_1}{F}$$

There can be
no x_2 terms
in G_1

$$G_2 = \frac{x_2}{F}$$

no x_1 terms
in G_2

$$\overset{D_1}{\boxed{[m_1 s^2 + (b_1 + b_2)s + k_1 + k_2]x_1 - \overset{N}{\boxed{(b_2 s + k_2)x_2}} = 0}}$$

$$\boxed{[m_2 s^2 + b_2 s + k_2]x_2 - (b_2 s + k_2)x_1 = F}$$

D_2

$$D_1 x_1 - N x_2 = 0$$

$$x_1 = \frac{N x_2}{D_1}$$

$$D_2 x_2 - N x_1 = F \rightarrow D_2 x_2 - \frac{N^2 x_2}{D_1} = F$$

$$D_1 \left(D_2 x_2 - \frac{N^2 x_2}{D_1} = F \right)$$

$$D_1 D_2 x_2 - N^2 x_2 = D_1 F$$

$$x_2 (D_1 D_2 - N^2) = D_1 F$$

$$G_{22} = \frac{x_2}{F} = \frac{D_1}{D_1 D_2 - N^2}$$

$$X_1 = \frac{N X_2}{D_1}$$

$$G_2 \Rightarrow \frac{X_2}{F} = \frac{D_1}{D_1 D_2 - N^2}$$

$$G_1 \Rightarrow \frac{X_1}{F} = \frac{X_1}{X_2} \cdot \frac{X_2}{F}$$

$$\frac{X_1}{X_2} = \frac{N}{D_1}$$

$$\uparrow$$

$$G_2$$

$$G_1 \Rightarrow \frac{X_1}{F} = \frac{X_1}{X_2} \cdot G_2 = \frac{N}{D_1} \cdot \frac{D_1}{D_1 D_2 - N^2} = \frac{N}{D_1 D_2 - N^2}$$

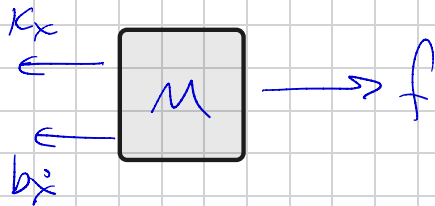
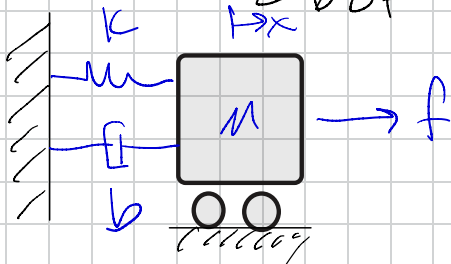
$$G_1 = \frac{N}{D_1 D_2 - N^2} = \frac{N}{(M_1 s^2 + (b_1 + b_2)s + K_1 + k_2) \cdot (M_2 s^2 + b_2 s + k_2)}$$

$$G_2 = \frac{D_1}{D_1 D_2 - N^2}$$

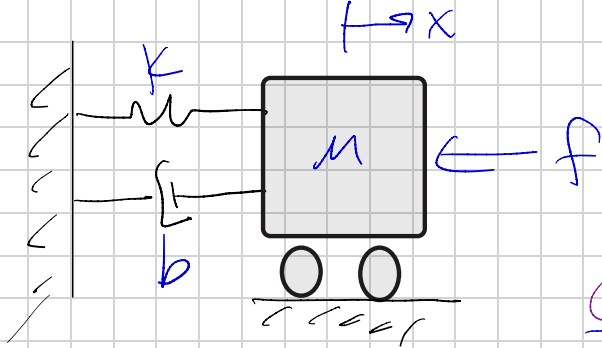
$$(M_2 s^2 + b_2 s + k_2) \rightarrow (b_2 s + k_2)^2$$

4th order

SDOF REVIEW

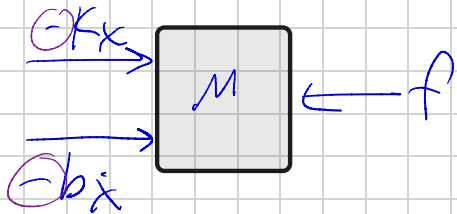


$$m\ddot{x} + b\dot{x} + kx = f$$



SDOF
Compression

Signs could get
weird



The mass has been displaced in the negative direction

Displacement input

